



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Automated Timetable Clash Detector and Resolver

Team ID: DSCPP-III-2026-T019

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T019
Semester: 3rd
Section: A, C, E, G
Domain: CSE

Team Members

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Mentor

Ms. Megha Garg

Interactions completed: _____ 2

Problem Statement & Motivation

Problem Statement

Colleges struggle with manual, error-prone timetable scheduling, leading to room, faculty, and section overlaps. Existing software often flags these conflicts but leaves staff to resolve them manually without explaining the underlying logic. This project addresses this gap by developing an automated C++ system that uses graph algorithms to actively detect, model, and resolve scheduling clashes without human intervention.

Motivation

- ▶ *Manual scheduling, missing auto-fix.*
- ▶ *Disrupted academic schedules.*
- ▶ *Passive flagging, opaque logic.*

Target Users / Stakeholders

- ▶ *College administrators.*
- ▶ *Students, faculty and institutional management.*
- ▶ *Automated clash-free scheduling.*

Objectives

Project Objectives

1. *Develop an automated timetable scheduling system.*
2. *Implement sweep-line clash detection and resolution.*
3. *Apply DSA and OOPS concepts in C++.*
4. *Test algorithm performance and system correctness.*
5. *Provide efficient and clash-free timetable output.*

Key Objective: *To enable fast, efficient, and automated resolution of college timetable clashes.*

Proposed Solution

Proposed Approach

1. *Add Timetable Data*
2. *Apply DSA Algorithms*
3. *Resolve Scheduling Clashes*
4. *Store Availability Records*
5. *Display Conflict-Free Output*

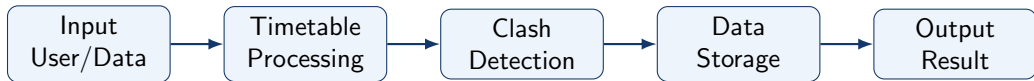
Key Features

- ▶ *Detect Schedule Clashes*
- ▶ *Auto-Resolve Conflicts*
- ▶ *Fast Hash Lookup*
- ▶ *Priority-Based Allocation*
- ▶ *View Resolved Timetable*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Arrays, Hash Tables, Graphs (DFS/BFS), Sorting</i>	<i>Clash Detection & Conflict Modeling</i>
OOPs with C++	<i>Classes, Objects, Structures, STL Vectors</i>	<i>Data Modeling & Auto-Resolution Engine</i>

System Architecture / Workflow



Major Components

- ▶ *Timetable Management*
- ▶ *Clash Detection & Analysis*
- ▶ *Clash Resolution*

Data / Control Flow

- ▶ *User enters timetable details*
- ▶ *System checks teacher, room section conflicts*
- ▶ *Available slots are identified and revised timetable is displayed*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++*
- ▶ Database: *PostgreSQL*
- ▶ Tools / Framework: *C++ STL*
- ▶ IDE: *VS Code*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Studied DSA concepts*
- ▶ *Identified core requirements*
- ▶ *Designed system architecture*
- ▶ *Implemented C++ models*
- ▶ *Prepared test datasets*

Individual Contribution

Member	Role	Contribution
M1	Lead	System design
M2	Developer	Core implementation
M3	DSA lead	Clash Detection
M4	Developer/Docs	Data modeling

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- ▶ *Working clash detection system*
- ▶ *Accurate conflict detection*
- ▶ *Clash resolution suggestions*
- ▶ *Efficient timetable management*

References

1. *C++ Documentation*
2. *DSA Textbook*
3. *C++ STL Reference*
4. *GitHub Repository*

Thank You
Questions & Suggestions